Loan Timing, Follow-on Transfers, and Playing Time

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1 Executive Summary

This document summarizes three follow-up analyses motivated by our previous discussion of loan-transfer sequences.

The three questions are:

1. Can we replace an arbitrary universal timing window with a more principled country-specific rule for identifying unusual loan endings?
2. What is actually contained in the previous broad Category C, "another move or re-loan"?
3. Can the transfer data be joined reliably to playing time, and what does the loan-versus-permanent comparison show?

The main findings are:

1. **Loan-ending calendars are strongly country-specific.** Most European domestic loans are concentrated around June 30, while England and Scotland are concentrated around May 31, Japan around January 31, and Argentina and Brazil around December 31.
2. **A reproducible country-calendar rule is feasible.** The proposed rule learns recurring ending dates outside the COVID-disrupted period and classifies each loan as calendar-normal, off-calendar, or unclassified. Among the 12,786 realised endings for which the necessary geography is available, 83.9% are calendar-normal and 16.1% are off-calendar. Outside the COVID window, 12.9% are off-calendar.
3. **Off-calendar loans look meaningfully different.** Their median duration is 170 days compared with 297 days for calendar-normal loans. They are much more likely to end in January, receive the existing early-termination label, and be followed by a new loan to a different club.
4. **The calendar rule should currently be treated as additional metadata, not a hard exclusion rule.** More than half of realised loan endings cannot currently be classified because the country is missing for one or both clubs in our dataset. We leave these cases unclassified rather than inferring the country.

5. **The previous Category C combines several economically different sequences.** Of 7,853 Category C cases, 5,976 are loans to a different club after returning to the lender, while only 331 are re-loans to the same borrower.

6. **The playing-time join works well within the covered leagues.** We identify 3,808 clean loan episodes and 11,350 clean permanent transfers across 14 long-covered European first divisions from 2012/13 to 2025/26.

7. **Raw, loan arrivals play more than permanent-transfer arrivals.** Our primary playing-time sample contains 3,808 loan arrivals and 11,350 permanent-transfer arrivals. After arriving at the receiving club, loanees play an average of 43.7% of the league minutes available to them, compared with 39.2% for players arriving through permanent transfers. The raw difference is therefore +4.5 percentage points.

8. **The raw difference largely disappears after accounting for observable differences between the players and receiving clubs.** The preferred regression uses the 13,865 episodes for which the required pre-transfer market-value data are available. After controlling for player value relative to the receiving squad, squad value, age, position, arrival timing, and league-by-season effects, the estimated loan–permanent difference is -0.8 pp, with a 95% confidence interval of $[-2.0, +0.4]$ pp. Thus, the data do not show a positive average loan association once these observed characteristics are taken into account.

9. **Player value relative to the receiving squad is particularly important.** Loanees tend to be more valuable relative to the clubs they join than permanent signings. Once this is taken into account, the positive raw loan association disappears.

10. **There is interesting heterogeneity.** Lower-relative-value loanees may play more than similar permanent signings, while England shows a robust negative loan association. These are exploratory follow-up findings rather than the main conclusion.

The key interpretation of the playing-time analysis is therefore:

> Loanees appear to play more in the raw data, but they are also systematically different from permanent signings. Once observable differences (especially player market value relative to the receiving club) are taken into account, there is no robust positive average loan-playing-time association.

2 Part I: Loan-Ending Timing and Country-Specific Calendars

2.1 Motivation

The previous loan-sequence analysis used timing windows such as 21 or 60 days after a loan return to identify potentially related follow-on movements.

The difficulty with such a rule is that the choice of 21 or 60 days is not naturally determined by the data.

For example, suppose the observed sequence is

$$A \to B \text{ on loan} \to A \text{ at the end of the loan} \to C.$$

Whether the final move is considered "closely related" depends mechanically on the chosen time window.

The analysis therefore asks:

> Can we identify a country's normal loan-ending calendar and treat returns away from those recurring dates as candidate edge cases?

2.2 Population

There are 27,402 realised loan endings in the current episode data.

They divide geographically as follows:

Table 1: Realised loan endings by geographic coverage

Type	N	Share
Domestic: both clubs mapped to same country	8,402	30.7%
Cross-border: both clubs mapped, different countries	4,989	18.2%
One or both clubs have unknown country	14,011	51.1%
Total	27,402	100.0%

The large unmapped group is important later: any country-calendar rule can only classify observations for which the relevant club countries are known.

2.3 Country-specific ending-date patterns

The ending-date distribution varies sharply across countries.

For domestic loans, the most common dates include:

Table 2: Selected country-specific loan-ending patterns

Country	Dominant date	Share of domestic endings
England	May 31	45.8%
Scotland	May 31	37.2%
Japan	January 31	86.9%
Argentina	December 31	62.5%
Brazil	December 31	56.9%
Italy	June 30	78.6%
Spain	June 30	81.7%
Germany	June 30	87.1%
France	June 30	92.4%
Netherlands	June 30	90.0%

Cross-border loans have a different pattern. Approximately 69.5% of all cross-border loan endings occur on June 30.

This is especially important for English clubs. May 31 is common for loans between two English clubs, but England-to-Europe loans overwhelmingly end on June 30.

For example:

Table 3: Selected English cross-border loans

Lender → borrower	Loans	Ending on borrower's usual date
England → Spain	157	128 on June 30
England → France	101	86 on June 30
England → Italy	97	75 on June 30
England → Germany	95	81 on June 30

This suggests that domestic country calendars and cross-border calendars should be treated separately.

2.4 COVID disruption

The period from March 1, 2020 through February 28, 2021 is visibly unusual.

Examples include:

- Italy: 89 domestic loan endings on August 31, 2020, compared with only 3 August 31 endings in all other years.
- Spain: substantial concentration around July 20–21, 2020.
- Brazil: 9 domestic endings on February 28, 2021 and none on that date in other years.

The rule therefore treats COVID as follows:

1. COVID-period observations are **excluded when learning** each country's normal calendar.
2. COVID-period observations are **not deleted**.
3. They are classified using the non-COVID calendar and separately flagged as COVID-period observations.

This avoids allowing extraordinary pandemic-era dates to become permanent "normal" dates.

2.5 Proposed country-calendar rule

The proposed primary rule is:

1. Use realised loan endings outside the COVID window.
2. Give a country its own domestic calendar only if it has at least 100 such observations.
3. Cluster ending dates that fall within ±3 days of one another.
4. The cluster centre is the most common observed date.
5. A cluster must:
   - recur in at least 3 calendar years; and
   - contain at least 2% of the country's endings.
6. Keep the largest clusters until either:
   - 90% of training endings are covered;
   - the next cluster is below the 2% threshold; or
   - 5 clusters have been retained.
7. Use one pooled calendar for cross-border loans.
8. If the relevant geography is unavailable, label the observation unclassified rather than guessing.

Formally, for a realised ending i,

$$CalendarStatus_i = \begin{cases} \text{normal}, & \text{if } d_i \text{ lies within } \pm 3 \text{ days of a retained date,} \\ \text{off-calendar}, & \text{if geography is known but no retained date matches,} \\ \text{unclassified}, & \text{if the required geography is unavailable.} \end{cases}$$

2.6 Calendars produced by the rule

Selected examples are:

Table 4: Selected calendars under the primary rule

Calendar	Retained dates	Normal coverage
England	May 31; Jan 1; Jun 30; Jan 5; Feb 1	70.4%
Scotland	May 31; Jan 1; May 5; May 1; May 14	70.5%
Japan	Jan 31; Dec 31	93.8%
Italy	Jun 30; Jan 30	91.1%
Spain	Jun 30	90.4%
Germany	Jun 30; Jan 30	92.2%
France	Jun 30	92.1%
Brazil	Dec 31; Nov 30; Aug 31; Jun 30; Jul 11	74.9%
Argentina	Dec 31; Jun 30	89.9%
Cross-border	Jun 30; Dec 31; Jan 30; May 31	89.7%

England, Scotland, and Brazil are substantially more dispersed than most other countries and hit the five-cluster cap before reaching the 90% target.

2.7 How many loans are classified as edge cases?

The primary rule yields:

Table 5: Calendar-rule classification

Classification	N	Share
Calendar-normal	10,732	83.9% of classifiable
Off-calendar	2,054	16.1% of classifiable
Geography / insufficient-country-data unclassified	14,616	53.3% of all realised endings

Outside the COVID window, the off-calendar share falls from 16.1% to 12.9%.

If England, Scotland, and Brazil—the three most dispersed calendars—are also set aside, the non-COVID off-calendar share is 9.5%.

2.8 Do off-calendar loans look different?

Yes.

Table 6: Calendar-normal versus off-calendar loans

Characteristic	Calendar-normal	Off-calendar
N	10,732	2,054
Median loan duration	297 days	170 days
Ends in January	6.5%	30.9%
Ends in July/August	0.3%	29.1%
Existing early-termination label	2.2%	25.9%
Next move within 1 day	23.4%	30.8%
Next move within 21 days	38.0%	45.9%
Next move within 60 days	58.9%	57.8%
Next move: permanent to borrower	19.3%	12.2%
Next move: loan to different club	35.4%	53.6%

The main distinction is therefore not whether the player eventually moves again within 60 days—that rate is almost identical.

Instead, off-calendar endings are:

- shorter;
- more heavily concentrated in January;
- more likely to receive the existing early-termination label;
- more likely to be followed quickly by a new loan elsewhere.

That is consistent with these observations being more likely to represent recalls, early endings, or restructuring.

It does **not** prove a particular contract clause.

2.9 Relation to the old 60-day rule

The two rules answer different questions.

Country-calendar rule:

Did the loan itself end on a normal date?

60-day follow-on rule:

Did another movement occur soon after the loan ended?

Among classifiable realised returns with the relevant follow-on information:

Table 7: Crossing the calendar rule with the 60-day follow-on rule

Group	N	Share	Interpretation
A: normal date, no follow-on $\leq$ 60d	4,499	35.2%	ordinary ending, no quick move
B: normal date, follow-on $\leq$ 60d	6,232	48.8%	normal ending, planned next move
C: off-calendar, no follow-on $\leq$ 60d	872	6.8%	unusual ending date
D: off-calendar, follow-on $\leq$ 60d	1,180	9.2%	strongest edge-case group

Group D is particularly interesting for future contract research because it combines:

$$\text{unusual ending date} + \text{quick subsequent movement.}$$

The recommendation is therefore to **retain both variables** rather than replace one with the other.

2.10 Recommended use of the timing rule

The calendar rule should currently be stored as **metadata**, not used as a hard exclusion rule.

Suggested fields include:

- `calendar_status`;
- `nearest_standard_date`;
- `days_from_standard_date`;
- `domestic_vs_crossborder`;
- `covid_flag`.

The main reason not to filter the dataset is that 53.3% of realised endings cannot currently be classified because of incomplete deterministic country mapping.

Dropping them would create a sample-selection problem driven by data coverage rather than football economics.

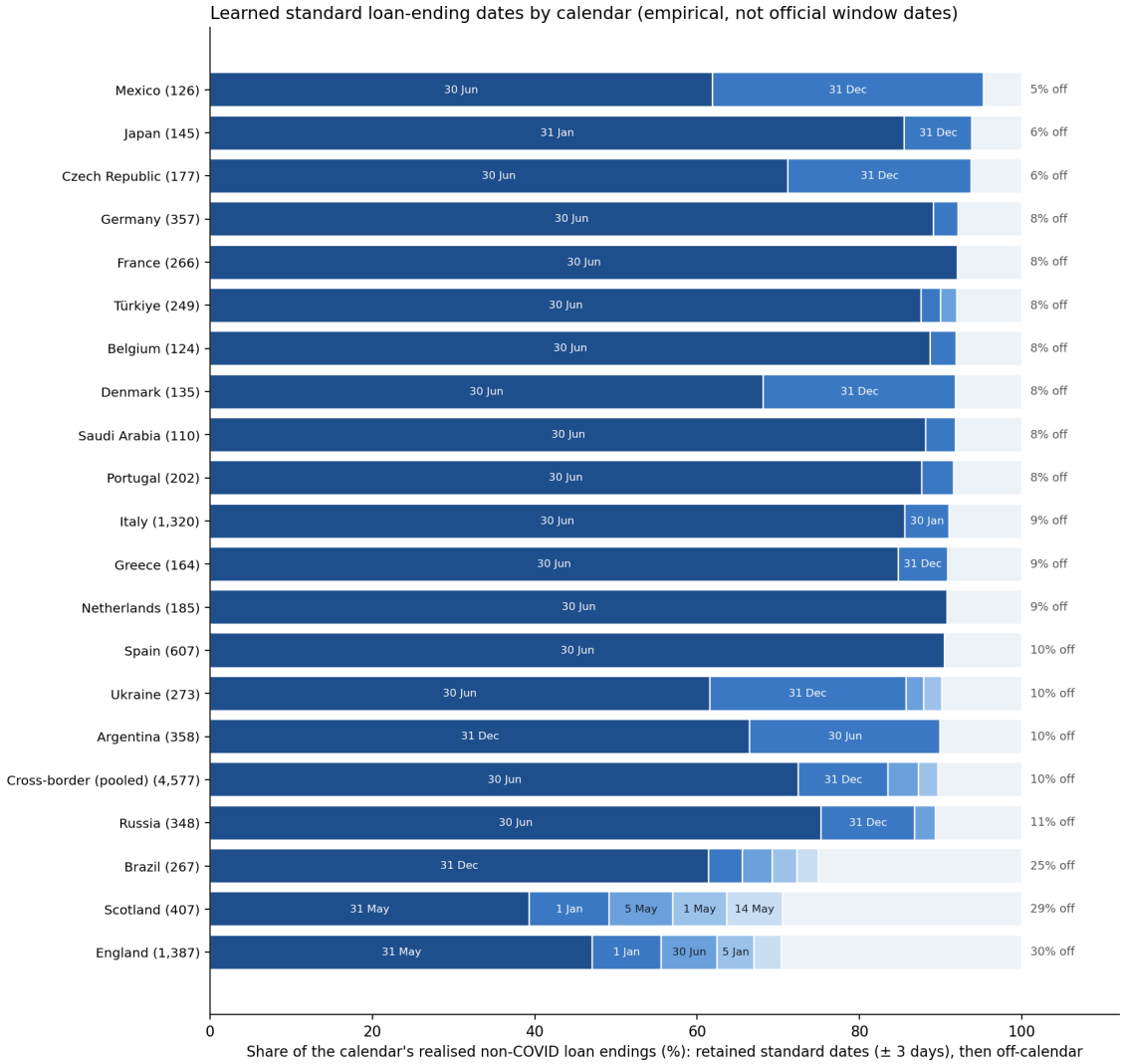


Figure 1: Learned standard loan-ending dates by country.

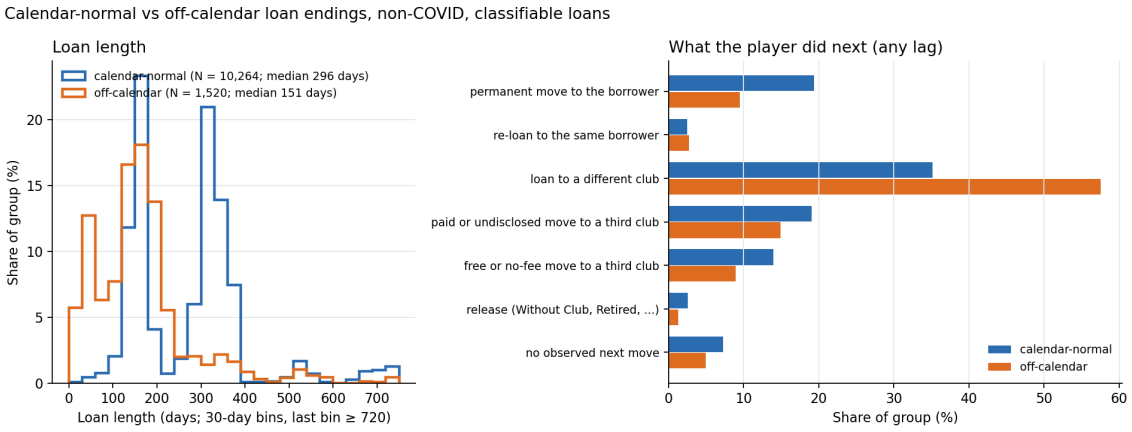


Figure 2: Comparison of calendar-normal and off-calendar loan endings.

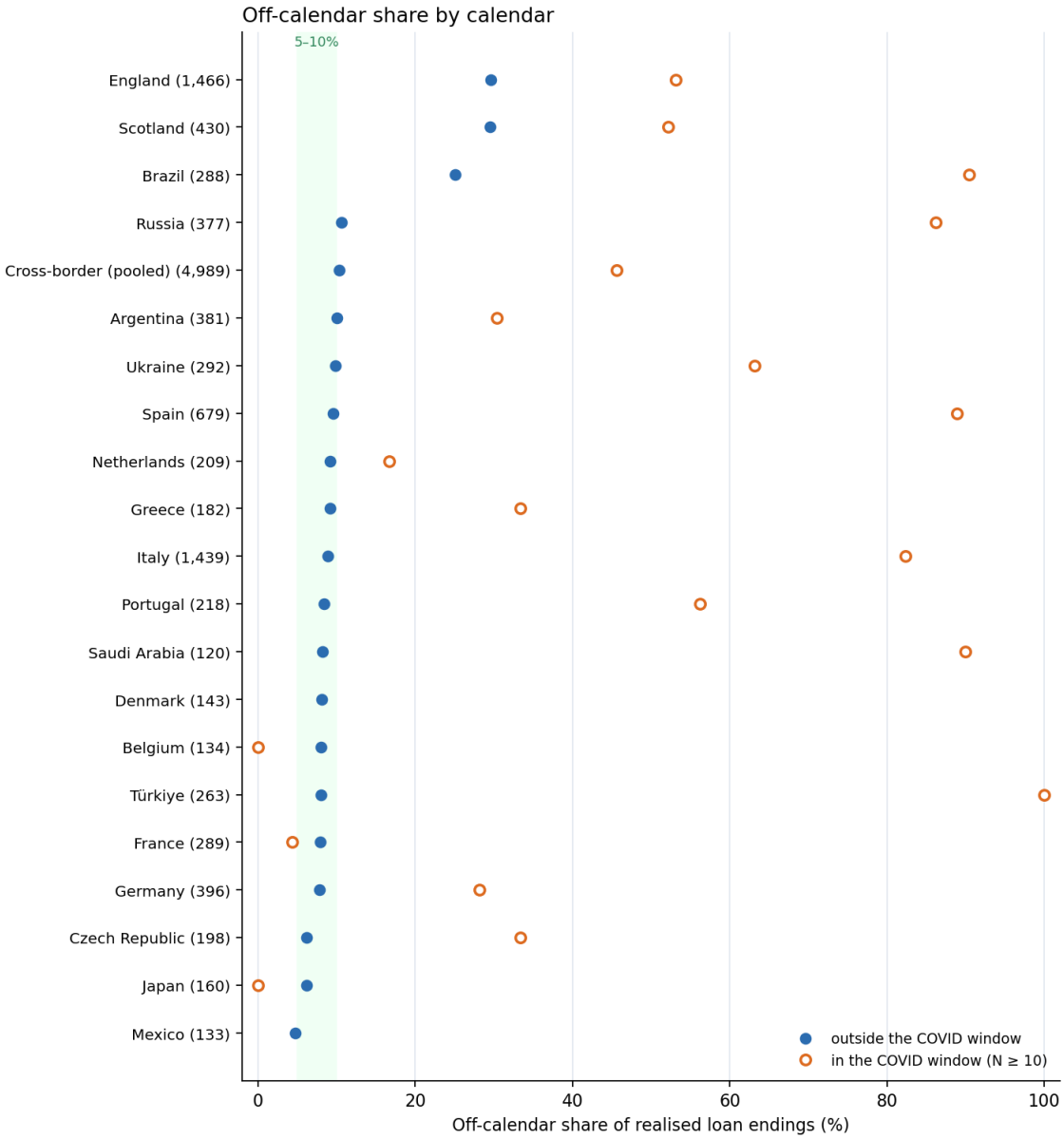


Figure 3: Off-calendar share by country under the proposed rule.

3 Part II: Decomposing Category C

3.1 Original classification

The earlier follow-on classification contained:

- B: permanent move to the borrower;
- C: another move or re-loan;

- D: sale to a third club.

Category C was too broad because it combined several economically different sequences. The question from the meeting was particularly whether a player:

$$A \to B \text{ on loan} \to A$$

was then:

$$A \to B \text{ again}$$

or instead:

$$A \to C.$$

3.2 Full decomposition

The 7,853 original Category C observations decompose as:

Table 8: Decomposition of original Category C

Subgroup	N	Within 1 day	Median lag
Re-loan to same borrower B	331	12%	31 days
Loan to different club C	5,976	32%	12 days
Free transfer to third club	1,090	52%	1 day
No-fee-shown move to third club	382	94%	1 day
Other	74	35%	8 days
Total	7,853		

The most important result is that the overwhelming majority of the re-loan category is:

$$A \to B \to A \to C,$$

not another loan to the same borrower.

Only 331 cases are:

$$A \to B \to A \to B$$

with the second move also being a loan.

3.3 Interpretation

The groups also differ in timing.

Same-borrower re-loans have a median lag of approximately 31 days after return.

Loans to a different club have a median lag of approximately 12 days.

Free transfers and no-fee-shown third-club moves are much more heavily concentrated immediately after the return.

The old Category C therefore mixed at least three different economic patterns:

1. extension / renewal of the relationship with the same borrower;
2. redirection to a different borrowing club;
3. permanent or administrative movement to a third club.

4 Part III: Joining Transfers to Playing Time

4.1 Coverage question

The first question was whether the transfer data can be connected reliably to actual playing time.

The underlying appearance data are sufficiently complete for 14 European first divisions over 2012/13–2025/26:

BE1, DK1, ES1, FR1, GB1, GR1, IT1, L1, NL1, PO1, RU1, SC1, TR1, UKR1.

Within those leagues, the join is highly successful.

Table 9: Playing-time join

Step	Loans	Permanent transfers
All candidate episodes	29,699	85,341
Receiving club in covered league-season	6,536	17,583
Reliable playing-time outcome	6,283	16,235
Primary analysis sample	3,808	11,350

Conditional on the receiving club being in a covered league-season:

96.1%

of loan episodes and

92.3%

of permanent-transfer episodes obtain a reliable playing-time outcome.

Thus, the main limitation is **league coverage**, not failed player or club matching.

(Playing time can be measured reliably for 6,283 loan episodes. For the main loan-versus-permanent comparison, we further restrict the sample to 3,808 loans that represent clean, comparable arrivals: the player must arrive with at least two-thirds of the season remaining, it must be his first spell at the receiving club, and the loan must have a recorded return to the lender. The remaining loans are not dropped because playing-time data are missing; they are excluded only from the primary comparison because they do not satisfy these comparability conditions.)

4.2 Definition of playing time

The primary outcome is the player's share of the receiving club's available league minutes:

$$PTShare_i = \frac{\text{player league minutes at receiving club}}{90 \times \text{receiving-club league matches in the observation window}}.$$

The window begins on the recorded transfer date and ends at the earlier of:

1. the receiving club's final league match of the season; or
2. the player's next recorded departure from the club.

This outcome is preferable to raw minutes because it adjusts for:

- mid-season arrivals;
- different league lengths;
- shortened seasons;
- early departures;
- different numbers of matches remaining.

Missing coverage is never interpreted as zero.

A true zero is recorded only when the receiving club's matches are sufficiently complete to establish that the player did not play.

4.3 Primary sample rules

The primary sample requires:

- reliable playing-time coverage;
- at least two-thirds of the receiving club's league season remaining at arrival;

- the player's first observed spell at the receiving club;
- for loans, a clean ending or scheduled ending with return to the lender.

This produces:

$$N_{\text{loan}} = 3,808$$

and

$$N_{\text{permanent}} = 11,350.$$

5 Part IV: Raw Playing-Time Comparison

5.1 Overall difference

The raw data initially suggest a substantial loan advantage.

Table 10: Raw playing-time comparison

	Loans	Permanent	Loan − permanent
N	3,808	11,350	
Mean share	0.437	0.392	+0.045
Median share	0.432	0.374	+0.058
p25	0.20	0.10	+0.10
p75	0.66	0.64	+0.02
p90	0.83	0.83	0.00
Exactly zero	4.1%	10.4%	−6.3 pp
Below 10%	15.0%	24.5%	−9.5 pp
Above 50%	42.8%	38.0%	+4.8 pp

The raw mean difference is therefore:

$$\boxed{PTShare_{\text{loan}} - PTShare_{\text{permanent}} = +4.5 \text{ pp}}$$

5.2 Where the raw difference occurs in the distribution

The raw difference is concentrated primarily in the lower part of the distribution.

Permanent signings are substantially more likely to:

- play zero league minutes;

- play fewer than 10% of available minutes.

At the upper end of the distribution, the groups are similar:

$$p90_{\text{loan}} = p90_{\text{permanent}} = 0.83.$$

Thus, the raw result is not simply that loan players dominate permanent players at every level.

Instead:

> Permanent signings contain a larger group of players who arrive and receive almost no league playing time.

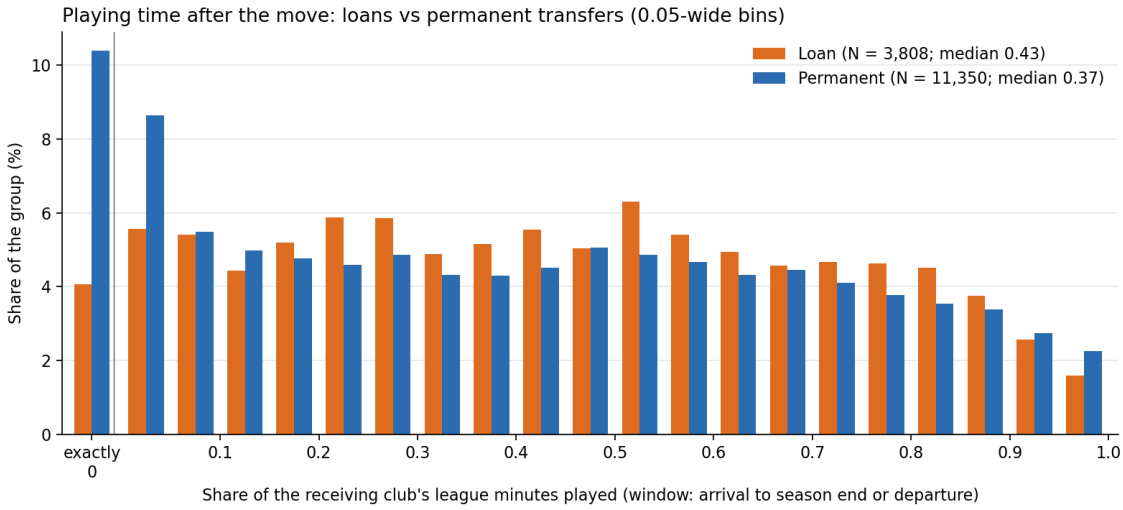


Figure 4: Distribution of playing-time shares for loans and permanent transfers.

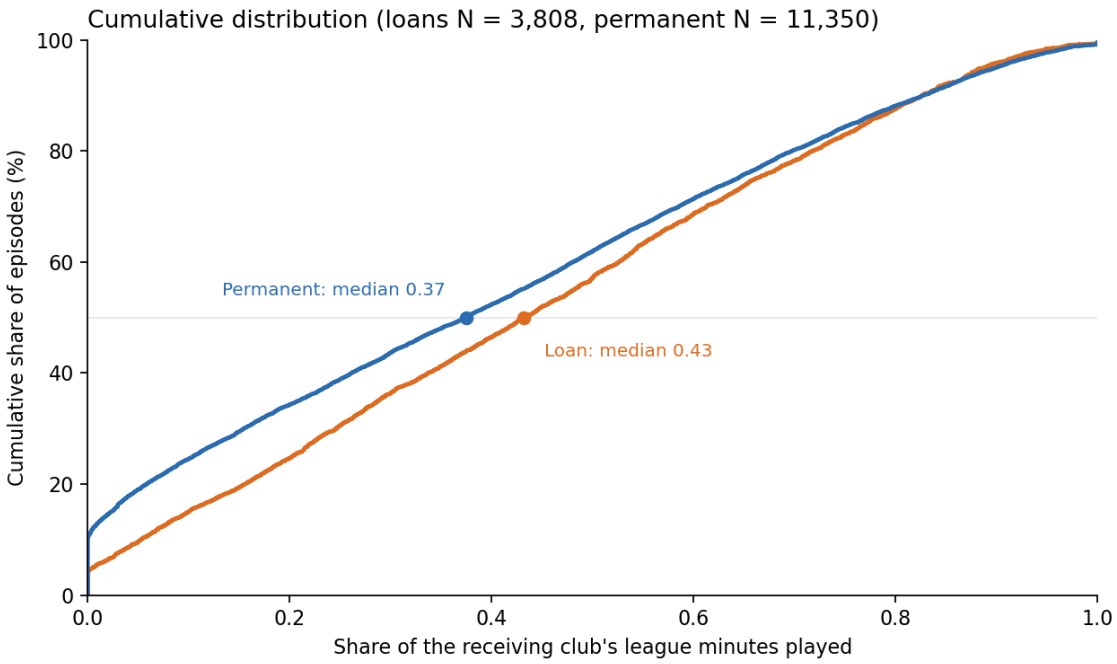


Figure 5: Empirical cumulative distribution of playing-time share.

5.3 Raw result by league

In the raw data, loanees play more on average in 13 of the 14 leagues.

England is the exception.

Selected raw differences include:

Table 11: Selected raw league differences

League	Loan mean	Permanent mean	Difference
Belgium	0.45	0.36	+0.091
Türkiye	0.50	0.40	+0.096
Scotland	0.48	0.42	+0.065
Italy	0.41	0.35	+0.053
France	0.45	0.39	+0.055
England	0.37	0.42	−0.047

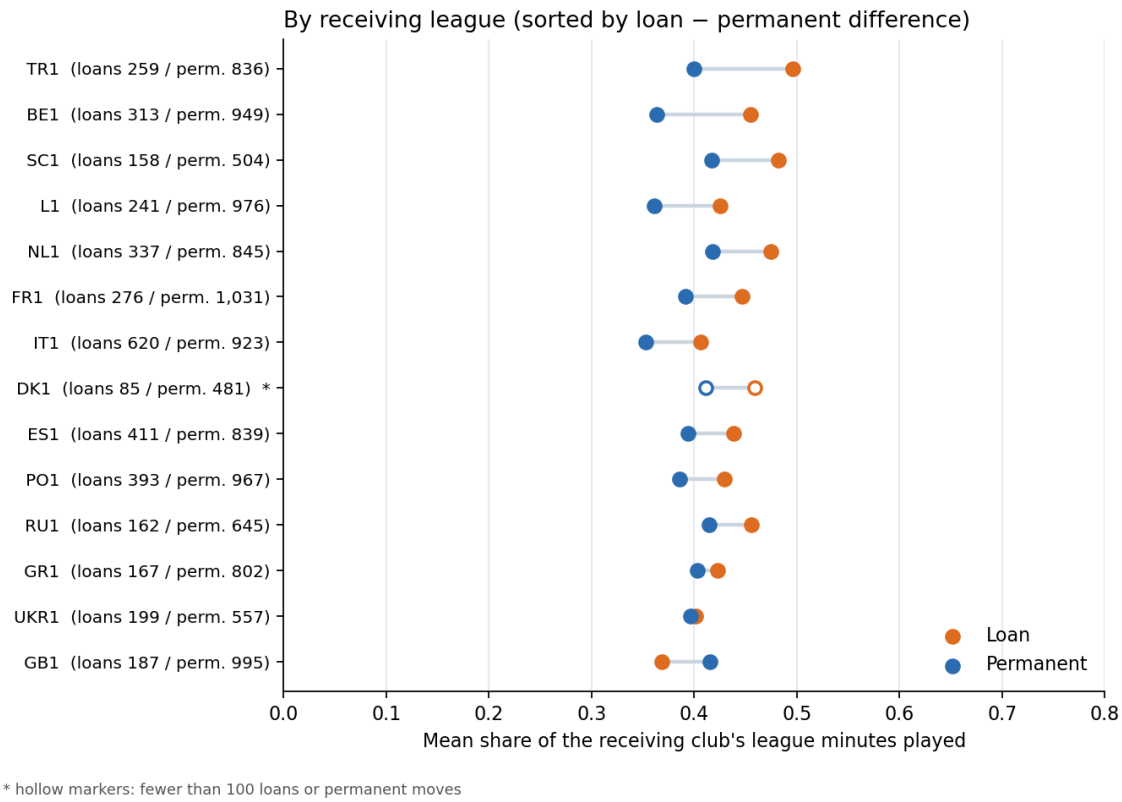


Figure 6: Raw mean playing-time share by receiving league.

5.4 Raw result by season

The mean playing-time share is higher for loans in all 14 arrival seasons.

Excluding the COVID seasons does not materially alter the result.

Table 12: COVID sensitivity of raw playing-time gap

Sample	Loan mean	Permanent mean	Difference
Primary sample	0.44	0.39	+0.045
Exclude 2019/20	0.44	0.39	+0.044
Exclude 2019/20 and 2020/21	0.44	0.39	+0.047

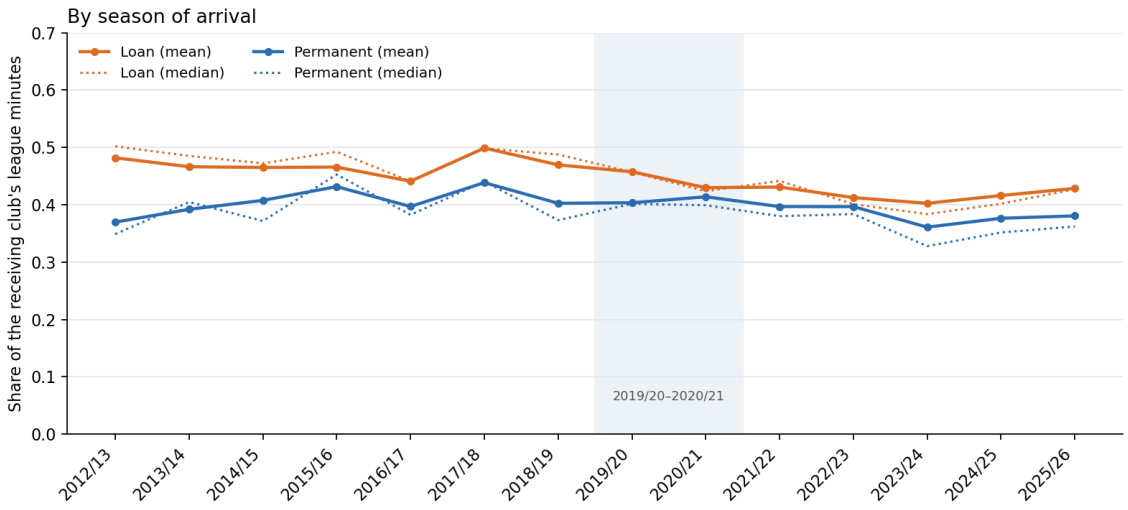


Figure 7: Raw playing-time shares by season of arrival.

6 Part V: Why the Raw Comparison Is Not Sufficient

6.1 Loans and permanent signings differ in observable characteristics

The two groups are not directly comparable.

Table 13: Selected characteristics before controls

Characteristic	Loans	Permanent
Mean age	23.4	24.1
Median player value	€2.0m	€1.2m
Median receiving-squad value	€47.8m	€57.9m
Median relative player value	3.9%	2.5%
Arrived before club's first league game	52.7%	78.5%
Zero league minutes	4.1%	10.4%

The largest standardized difference is in relative player value.

6.2 Relative player value

Define:

$$RelativeValue_i = \frac{\text{player market value before move}}{\text{receiving-club squad value before move}}.$$

This directly implements the idea that the same nominal player value can mean very different things at clubs of different strength.

For example:

$$\frac{\text{10m player}}{\text{50m squad}} = 20\%$$

whereas

$$\frac{\text{10m player}}{\text{500m squad}} = 2\%.$$

The former player is much more important relative to the club's squad.

The median relative value is:

$$3.9\%$$

for loanees and

$$2.5\%$$

for permanent signings.

Thus, loanees tend to arrive as relatively more valuable players for the clubs receiving them.

7 Part VI: Controlled Playing-Time Analysis

7.1 Regression framework

The primary linear model is of the form:

$$PTShare_i = \alpha + \beta Loan_i + X_i'\gamma + \delta_{\ell(i),s(i)} + \varepsilon_i,$$

where:

- $PTShare_i$ is the player's share of available receiving-club league minutes;
- $Loan_i$ equals 1 for a loan and 0 for a permanent transfer;
- X_i contains observable player, club, and timing controls;
- $\delta_{\ell(i),s(i)}$ are league-by-season fixed effects.

The preferred controls include:

- arrival timing;

- age;
- age squared;
- broad playing position;
- log relative player value;
- log receiving-squad value;
- league-by-season fixed effects.

The coefficient β is interpreted as an **adjusted association**, not a causal treatment effect.

7.2 Regression sequence

Table 14: Loan–permanent difference across regression specifications

Specification	N	Loan coefficient	95% CI
Raw, full sample	15,158	+4.5 pp	[+3.4,+5.7]
Raw, market-value sample	13,865	+2.4 pp	[+1.3,+3.5]
+ league × season, timing	13,865	+2.6 pp	[+1.4,+3.8]
+ age, age^2, position	13,865	+4.2 pp	[+2.9,+5.4]
+ relative value, squad value (preferred)	13,865	−0.8 pp	[−2.0, +0.4]
Club × season FE alternative	13,443	−2.1 pp	[−3.5, −0.7]

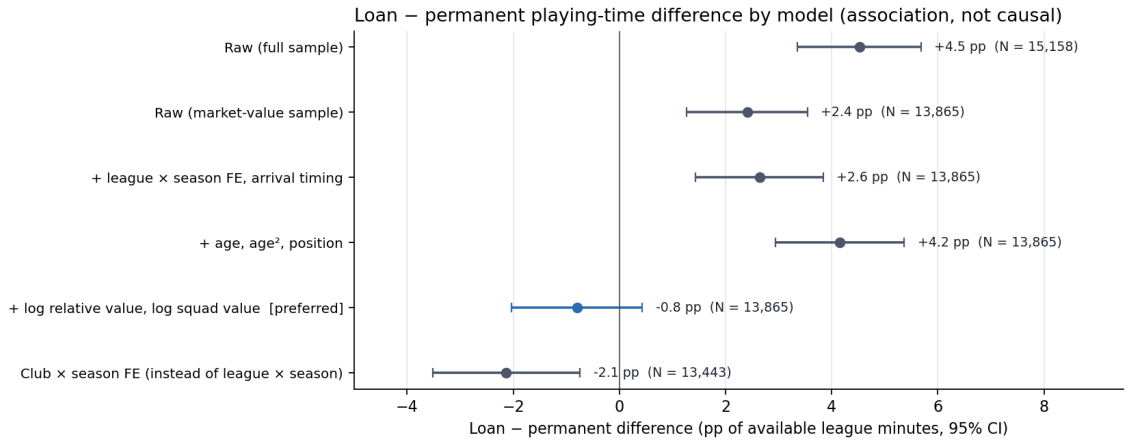


Figure 8: Loan–permanent playing-time difference across regression specifications.

7.3 Sample selection versus covariate adjustment

The movement from:

+4.5 pp

to:

$$-0.8 \text{ pp}$$

is a total change of 5.3 percentage points.

It has two components.

1. Sample-selection effect

Requiring a usable pre-move valuation changes:

$$+4.5 \rightarrow +2.4.$$

This accounts for approximately:

$$2.1/5.3 = 39.7\%$$

of the total movement.

The dropped permanent signings are unusually low-playing observations:

$$\text{mean playing-time share} = 0.17$$

and

$$31.1\%$$

of them record zero league minutes.

Thus, part of the raw loan advantage disappears simply because the regression sample differs from the full sample.

2. Covariate-adjustment effect

Holding the market-value sample fixed, the coefficient moves:

$$+2.4 \rightarrow -0.8.$$

This is a 3.2 percentage-point adjustment.

The market-value variables are responsible for almost all of the final downward movement.

Age and position actually push the loan coefficient upward before market value is introduced.

7.4 Preferred result

The preferred specification yields:

$$\boxed{\hat{\beta}_{Loan} = -0.8 \text{ pp}}$$

with:

$$95\% \text{ CI} = [-2.0, +0.4] \text{ pp.}$$

The estimate is therefore statistically indistinguishable from zero.

The defensible interpretation is:

> Conditional on observed player value relative to the receiving squad, squad value, age, position, arrival timing, and league-season, loan arrivals play approximately the same amount as permanent arrivals on average.

A slightly more cautious formulation is:

> There is no robust positive loan association after controls; if anything, the remaining adjusted association is small and negative.

7.5 Club-by-season fixed effects

An alternative model compares loan and permanent arrivals entering essentially the same club environment in the same season.

The estimate is:

$$\boxed{-2.1 \text{ pp}}$$

with:

$$95\% \text{ CI} = [-3.5, -0.7] \text{ pp.}$$

There are 1,623 club-seasons containing both loan and permanent arrivals, providing substantial within-club-season identifying variation.

This strengthens the conclusion that the positive raw association does not survive more comparable within-environment comparisons.

8 Part VII: Why Relative Player Value Matters

In the preferred model, relative player value is strongly associated with playing time.

A doubling of player value relative to receiving-squad value is associated with approximately:

$$+6.1 \text{ pp}$$

more playing time.

This helps explain the raw loan advantage because loanees are systematically more valuable relative to their destination squads.

8.1 Interaction between loan status and relative value

The loan association varies substantially across the relative-value distribution.

Table 15: Adjusted loan–permanent difference by relative-value fifth

Fifth	Median relative value	Loans	Permanent	Difference
Q1	0.5%	342	2,431	+8.0 pp
Q2	1.6%	634	2,139	+0.1 pp
Q3	2.8%	739	2,034	−1.7 pp
Q4	4.7%	851	1,922	−1.4 pp
Q5	9.1%	1,095	1,678	−2.4 pp

The positive adjusted loan difference is concentrated in the lowest fifth.

The estimated interaction crosses roughly around:

$$2.3\%$$

of receiving-squad value.

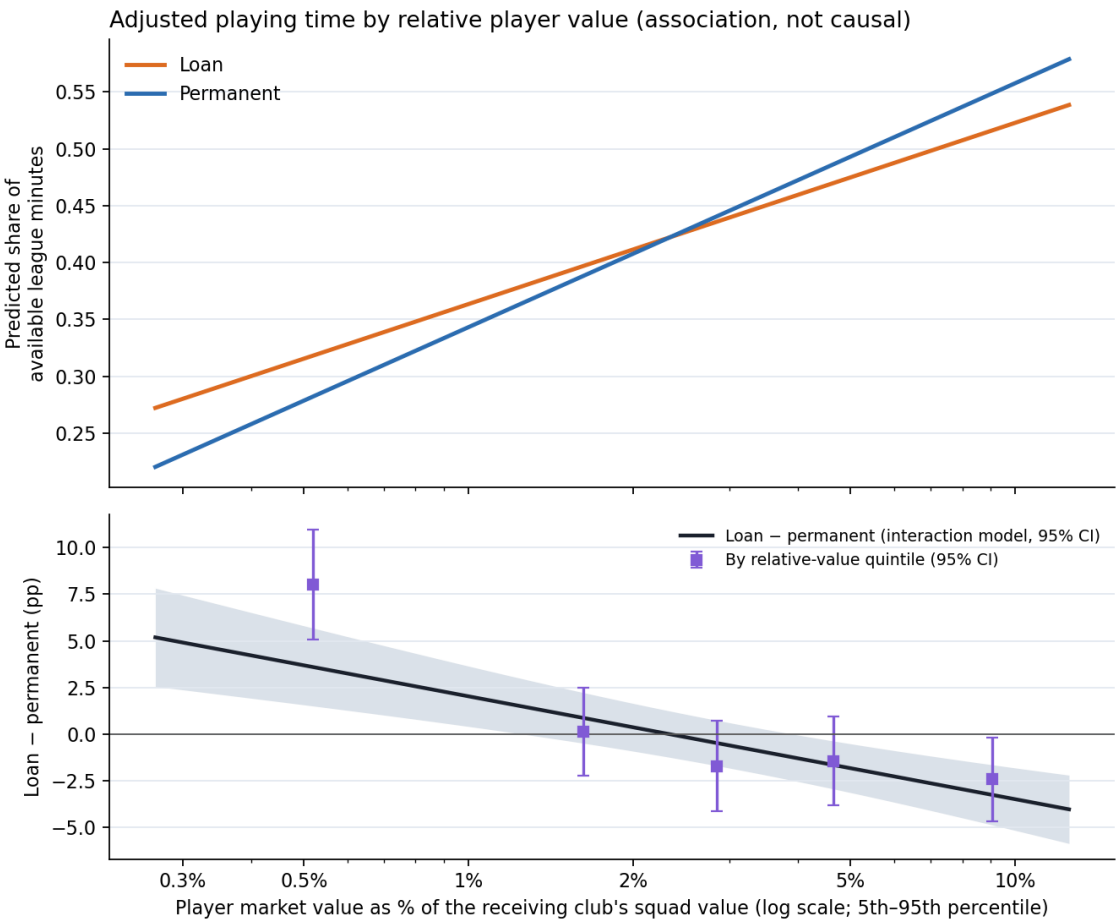


Figure 9: Adjusted playing time by relative player value and loan status.

8.2 Interpretation of the low-relative-value result

This result is potentially interesting economically.

One possible interpretation is that low-relative-value loanees are being acquired under a different development or usage model than low-relative-value permanent signings.

However, this is not yet a main conclusion.

After adding prior playing time on the smaller subsample, the Q1 estimate is:

$$+6.2 \text{ pp}$$

but this is based on only:

$$76$$

loan observations.

The confidence interval is wide.

The result therefore remains:

exploratory

9 Part VIII: Zero-Minute Players

Permanent signings are much more likely to receive no league playing time.

In the full primary sample:

$$P(PTShare = 0 \mid Loan) = 4.1\%$$

versus

$$P(PTShare = 0 \mid Permanent) = 10.4\%.$$

Dropping zero-minute observations reduces the raw difference from:

$$+4.5 \text{ pp}$$

to approximately:

$$+1.8 \text{ pp}.$$

Thus, the zero-minute mass explains a substantial part of the raw loan advantage.

However, after controls:

- excluding exact zero-minute players gives -1.3 pp;
- the probability of zero minutes is about 1.4 pp lower for loans;
- fractional-logit estimation of the playing-time share gives approximately -0.4 pp.

The conclusion therefore does not depend on treating the bounded outcome as a simple OLS variable.

10 Part IX: Prior Playing-Time Robustness

A remaining selection concern is that loanees and permanent signings may differ in playing ability or usage before moving.

Prior playing time can be measured for a selected subsample where the origin club also has sufficient historical appearance coverage.

The measurable subsample contains:

$$1,514$$

loans and

$$4,163$$

permanent transfers.

The groups differ substantially before the move.

Mean prior playing-time share:

$$0.30$$

for loanees and

$$0.56$$

for permanent signings.

On the same 5,677 observations:

Table 16: Prior-playing-time robustness

Specification	Loan coefficient	95% CI
Preferred controls, no prior PT	−3.4 pp	[−5.1, −1.6]
Preferred controls + prior PT	−1.0 pp	[−2.8, +0.8]

Controlling for prior playing time therefore moves the estimate back toward zero.

Importantly:

$$-1.0 \text{ pp}$$

is very close to the full preferred estimate:

$$-0.8 \text{ pp}.$$

This provides an independent robustness check supporting the main interpretation.

The prior-playing-time model is not used as the primary specification because the subset is selected toward players coming from covered first divisions and who spent the relevant prior period at their origin club.

11 Part X: England as an Exception

England is the clearest league-specific exception.

Table 17: England robustness

Specification	Loan–permanent difference
Raw full sample	−4.7 pp
Preferred adjusted model	−8.9 pp
Club × season FE	−10.6 pp
Prior-PT subsample, no prior control	−9.1 pp
Prior-PT subsample + prior PT	−6.8 pp

The negative England result is not driven by one obvious season or club.

Dropping one season at a time keeps the estimate between approximately:

$$-10.2$$

and

$$-7.6 \text{ pp.}$$

Dropping one receiving club at a time keeps it between approximately:

$$-9.8$$

and

$$-8.0 \text{ pp.}$$

The result is also negative across position and arrival-timing robustness checks.

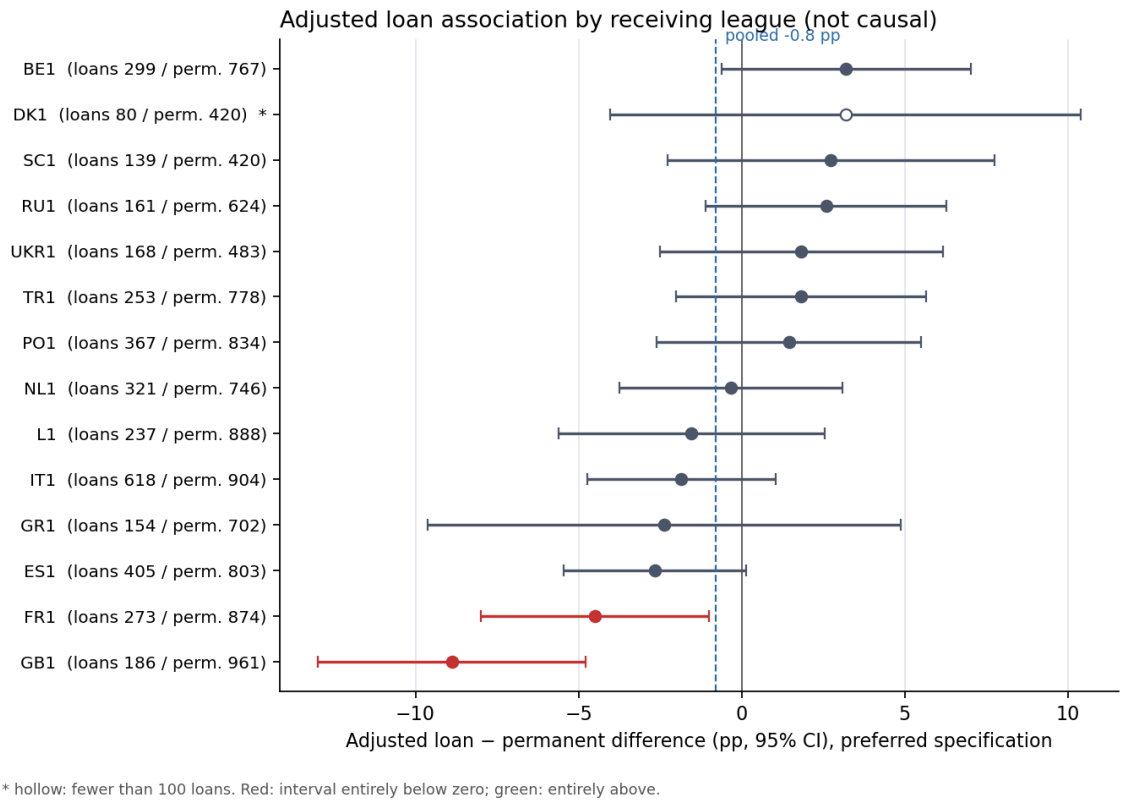


Figure 10: Adjusted loan–permanent playing-time difference by receiving league.

France is also significantly negative in the adjusted league-specific model.

No league has a confidence interval lying entirely above zero after controls.

The England result is therefore a credible descriptive heterogeneity finding, but the mechanism remains unknown.

12 Part XI: Valuation Timing and Other Robustness Checks

The market values used in the controlled analysis are strictly dated before the move.

Approximately:

- 79% of loan valuations;
- 81% of permanent-transfer valuations

are within 90 days of the move.

Only:

1.5%

of loan valuations and

$$2.1\%$$

of permanent-transfer valuations are older than one year.

Restricting the sample to fresher valuations produces:

Table 18: Valuation-age sensitivity

Valuation restriction	Loan coefficient	95% CI
All valuations	-0.8 pp	$[-2.0, +0.4]$
Within 180 days	-1.1 pp	$[-2.4, +0.1]$
Within 365 days	-1.0 pp	$[-2.2, +0.2]$

Stale valuations do not appear to drive the result.

13 Part XII: Summary of Core Playing-Time Specifications

Table 19: Core specifications and robustness checks

Model	N	Loan coefficient	95% CI
Raw OLS, full primary sample	15,158	+4.5 pp	[+3.4,+5.7]
Raw OLS, market-value sample	13,865	+2.4 pp	[+1.3,+3.5]
League × season FE + timing	13,865	+2.6 pp	[+1.4,+3.8]
+ age, age^2, position	13,865	+4.2 pp	[+2.9,+5.4]
Preferred + market values	13,865	-0.8 pp	$[-2.0, +0.4]$
Club × season FE	13,443	-2.1 pp	$[-3.5, -0.7]$
Preferred, nonzero players	12,906	-1.3 pp	$[-2.6, -0.0]$
Prior-PT subsample, no prior PT	5,677	-3.4 pp	$[-5.1, -1.6]$
Prior-PT subsample + prior PT	5,677	-1.0 pp	$[-2.8, +0.8]$

The central pattern is clear:

The positive raw loan advantage does not survive the controls.

Across the main controlled specifications, the remaining association is approximately zero to slightly negative.

14 Part XIII: Interpretation

14.1 What the evidence supports

The analysis supports the following statements:

1. Loanees play more than permanent signings in the raw sample.
2. Much of that difference comes from permanent signings with very low or zero playing time.
3. Loanees are systematically more valuable relative to the receiving squads they join.
4. After observed player, club, timing, and league-season differences are controlled for, there is no positive average loan-playing-time association.
5. Lower-relative-value loanees may behave differently from other players.
6. England has a robustly negative adjusted loan association.

14.2 What the evidence does not support

The analysis does **not** identify the causal effect of a loan contract.

Loan status is not randomly assigned.

Unobserved factors may affect both:

$$\text{loan vs permanent acquisition}$$

and

$$\text{subsequent playing time.}$$

Potential omitted factors include:

- development plans;
- contract length;
- wage structure;
- promises or expectations of playing time;
- tactical fit;
- injury risk;
- the lender's plans;
- information known privately by the player and clubs.

Therefore the preferred wording is:

> Conditional on the observable player, club, timing, and league-season characteristics available in the data, there is no positive average association between arriving on loan and subsequent league playing time relative to permanent transfers.

15 Part XIV: Current Decisions and Open Questions

15.1 Timing rule

Current recommendation:

- retain all loan observations;
- attach country-calendar status as metadata;
- do not hard-filter off-calendar observations;
- flag COVID separately;
- prioritize off-calendar + quick-follow-on cases for deeper contract research.

Professor Freund can decide later whether a particular empirical analysis should:

- exclude off-calendar observations;
- analyze them separately;
- use only main calendar dates;
- retain all observations and use calendar status as a covariate.

15.2 Category C

The classification issue is largely resolved.

The remaining limitation is statistical power for same-borrower re-loans in the playing-time sample.

A larger or broader appearance dataset would be required for a strong analysis of that specific subgroup.

15.3 Playing time

The main playing-time analysis is sufficiently validated for presentation.

Potential follow-up questions, depending on Professor Freund's interest, include:

1. Why is England different?
2. Why do low-relative-value loanees appear to receive more playing time?
3. Do specific loan contract clauses correlate with playing time?
4. Does the country-calendar edge-case classification interact with playing-time outcomes?
5. Can injury data improve the comparison?
6. Can the playing-time universe be extended to lower divisions or additional leagues?

These are follow-up research questions rather than necessary checks before presenting the current results.

16 Conclusion

The meeting follow-up produced three main advances.

First, loan-return timing can be described more systematically using country-specific empirical calendars rather than a single arbitrary universal date or follow-on window. The resulting off-calendar observations form a smaller and substantively distinct group that appears more consistent with early termination or rapid redirection.

Second, the broad Category C has been decomposed into economically different follow-on sequences, with different-club re-loans accounting for most of the category and same-borrower re-loans representing a relatively small subset.

Third, the transfer data can be joined reliably to playing time in 14 long-covered European first divisions. The raw data show a 4.5 percentage-point playing-time advantage for loanees, but that advantage does not survive controls for observed player and club characteristics, particularly player value relative to the destination squad. The preferred estimate is close to zero, and the same conclusion survives multiple robustness checks, including prior playing time.

The main empirical takeaway is therefore:

> The raw data make loanees look more heavily used than permanent signings, but this largely reflects which players are loaned and which clubs receive them. Among observably comparable arrivals, there is no robust positive average loan-playing-time association.

The strongest remaining heterogeneity is in England and among low-relative-value players, both of which provide natural directions for future analysis.